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A FAIRNESS AND BIAS AUDIT SYSTEM PROPOSAL REPORT

ON

BIASAPERTURE: A DIAGNOSTIC AND EVALUATIVE FRAMEWORK FOR AUDITING DEMOGRAPHIC BIAS IN FACIAL ANALYSIS SYSTEMS

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ABSTRACT

This proposal describes BiasAperture, a diagnostic framework for evaluating demographic disparities in facial-analysis model predictions and producing traceable research evidence. It specifies five modules for data ingestion, model interfacing, fairness metrics, surrogate attribution, and report generation, and discusses four metrics: Demographic Parity Difference, Equalized Odds Difference, Equal Opportunity Difference, and Disparate Impact Ratio. Fairlearn and AIF360 are proposed as separate metric implementations; their use does not constitute independent assessment. The reviewed implementation computes subgroup confidence intervals with an empirical percentile bootstrap that defaults to $B = 200$; adapters do not pass a larger resample budget. This does not meet the proposed NFR-002 minimum of 1,000 resamples. A separate global-summary path uses a BCa bootstrap routine, which does not close the subgroup-interval gap. The demographic logistic surrogate summarizes associations between demographic indicators and model predictions; it does not explain image pixels, identify causes or confounders, or decompose a disparity. UTKFace was excluded from the completed scope; FairFace is the retained benchmark. The proposed report crosswalk is a research aid, not evidence of regulatory compliance or a compliance determination. BiasAperture is diagnostic in scope and does not modify model weights, apply mitigations, or generate demographic images.

Keywords: Algorithmic Fairness, Bias Auditing, Demographic Parity, Disparate Impact, EU AI Act, Facial Analysis, Fairness Metrics, NIST AI RMF, Surrogate Attribution

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
CLI	Command-Line Interface
CPU	Central Processing Unit
CSV	Comma-Separated Values
DIR	Disparate Impact Ratio
DPD	Demographic Parity Difference
EOD	Equalized Odds Difference
EOP	Equal Opportunity Difference
EU	European Union
GPU	Graphics Processing Unit
HTML	HyperText Markup Language
ITA	Individual Typology Angle
JSON	JavaScript Object Notation
NIST	National Institute of Standards and Technology
PDF	Portable Document Format
RAM	Random-Access Memory
RMF	Risk Management Framework
SHAP	SHapley Additive exPlanations (image-based spatial explanations deferred)
SOLID	Single Responsibility, Open-Closed, Liskov Substitution, Interface Segregation, Dependency Inversion
UI	User Interface
VRAM	Video Random-Access Memory

LIST OF SYMBOLS

α	Significance threshold for chi-squared testing of independence between the predicted-label distribution and the demographic subgroup ($\alpha = 0.05$, NFR-001)
n	Subgroup sample size; the minimum threshold below which a metric is flagged as statistically insufficient rather than reported ($n \geq 30$, NFR-003)

1. INTRODUCTION

1.1. Background

Automated facial analysis systems, covering perceived gender classification, age estimation, and race or ethnicity prediction, are now embedded across commercial and public-sector applications ranging from photo tagging and demographic analytics to surveillance and access control. As these systems have proliferated, a substantial body of empirical research has documented that many of them exhibit statistically significant accuracy disparities across demographic subgroups, particularly by skin tone, gender presentation, and age. The foundational audit by Buolamwini and Gebru demonstrated that leading commercial gender-classification services performed markedly worse on darker-skinned female subjects than on lighter-skinned male subjects, with error rates as high as 34.7% for the worst-performing intersectional subgroup against as low as 0.8% for the best-performing one [1]. That finding catalysed a wider research agenda into fairness auditing for computer vision, subsequently surveyed comprehensively by Dehdashtian et al., who catalogue the sources of bias that arise across the machine learning pipeline, from data collection and annotation through model training and deployment, and the corresponding families of fairness metrics and mitigation techniques proposed in response [2].

Despite this growing awareness, the computer-vision literature contains comparatively few reusable workflows for systematic, reproducible audits of facial-analysis models. Dehdashtian et al.'s survey describes a substantial body of fairness metrics and mitigation techniques, while noting less work on standardising the auditing workflow itself [2].

The regulatory context is relevant but does not make an audit report a compliance determination. Regulation (EU) 2026/1744 amended the application timetable for high-risk-system requirements under Regulation (EU) 2024/1689: the relevant Chapter III requirements apply from 2 December 2027 for Annex III systems and 2 August 2028 for Annex I systems. Article 10 has no single, independent application date; the applicable date depends on the system category and the amended timetable [3, 4]. The NIST AI Risk Management Framework (AI RMF) is a complementary voluntary framework, organised around Govern, Map, Measure, and Manage [5].

1.2. Problem Statement

This project addresses the engineering challenge of combining fairness metrics, benchmark datasets, and statistical analysis in a reusable workflow for facial-analysis systems. Existing studies, including Gender Shades, evaluate particular systems and datasets; a repeatable software workflow could help researchers conduct comparable

evaluations. Such outputs can contribute traceable research evidence, but do not by themselves satisfy regulatory obligations or demonstrate compliance.

1.3. Objectives

1.3.1. General Objective

To design and propose a diagnostic software platform, BiasAperture, for evaluating demographic accuracy disparities in third-party facial-analysis models and reporting traceable research evidence in a standardised format.

1.3.2. Specific Objectives

1. To design a modular software architecture that accepts a trained facial analysis model, or a file of its precomputed predictions, together with a demographically labelled benchmark dataset, and computes subgroup and intersectional fairness metrics from it.
2. To specify the use of AIF360 and Fairlearn as separate implementations of fairness metrics, alongside statistical testing, and to document the limitations of those measures and their uncertainty.
3. To generate a standardised, exportable fairness report, structurally informed by the Model Cards [6] and Datasheets for Datasets [7] documentation conventions, that communicates subgroup performance disparities in a form accessible to both technical and non-technical stakeholders.
4. To specify a case study using a publicly available, demographically annotated benchmark; FairFace is the retained dataset [8], and UTKFace was excluded from completed scope.
5. To propose a crosswalk from fairness metrics to relevant provisions of Article 10 and Annex IV of the European Union (EU) Artificial Intelligence (AI) Act and to relevant outcomes in the National Institute of Standards and Technology (NIST) AI Risk Management Framework (RMF), as research traceability rather than proof of compliance.

1.4. Scope and Limitations

1.4.1. Scope

The proposal scopes BiasAperture as a diagnostic and evaluative platform: (i) subgroup and intersectional evaluation of predictions from a facial-analysis classifier; (ii) four disparity metrics, Demographic Parity Difference (DPD), Equalized Odds Difference (EOD), Equal Opportunity Difference (EOP), and Disparate Impact Ratio (DIR), with statistical tests and confidence intervals; (iii) demographic-dummy

logistic-surrogate attribution, which describes associations between demographic indicators and model predictions rather than explaining image content or identifying causal or proxy-confounding mechanisms; and (iv) a proposed Model-Cards- and Datasheets-informed HyperText Markup Language (HTML) report with a regulatory crosswalk presented as research traceability. FairFace is the retained benchmark [8]; UTKFace was excluded from the completed scope.

1.4.2. Limitations

BiasAperture does not mitigate disparities, retrain models, or generate synthetic demographic data. Results depend on benchmark-label quality; FairFace is the retained dataset and UTKFace was excluded from completed scope. The subgroup interval implementation currently uses a percentile bootstrap with default $B = 200$; it does not meet NFR-002's planned minimum of 1,000 resamples, notwithstanding the separate global-row BCa implementation. The demographic logistic surrogate uses demographic indicators and model predictions, so it cannot explain pixels, establish causation, identify proxy confounders, or decompose observed disparities. The planned report is intended to provide traceable research evidence, not legal advice, a regulator-ready dossier, or proof of compliance. Runtime and other performance values in this report are targets unless identified as measured.

2. LITERATURE REVIEW

Chapter 1 described the established literature on algorithmic fairness and the engineering challenge of applying it in computer vision audits. This chapter reviews selected scientific, mathematical, and regulatory sources that inform the proposed BiasAperture architecture.

Rather than treating the literature as an uncurated catalogue of citations, we organize this review across seven coherent thematic pillars:

1. **The Empirical Mandate**: Foundational evidence demonstrating persistent demographic and intersectional disparities in facial analysis systems;
2. **Mathematical Formulations of Fairness**: Formal definitions of error-rate and selection-rate parities, along with critiques of borrowed legal thresholds;
3. **Statistical Rigor and Demographic Inference Epistemology**: Uncertainty quantification, sample-size guards, and the mathematical bounds imposed by demographic inference accuracy;
4. **Explainability Foundations and Attribution Bounds**: Cooperative game theory, additive feature attribution, and impossibility theorems bounding post-hoc explanations;
5. **Standardized Documentation and Regulatory Governance**: Recognized reporting conventions and legal-to-technical verification mappings;
6. **The Pipeline and Workflow Void**: The structural gap between theoretical metrics and reusable, air-gapped auditing software; and
7. **Frontiers and Future Diagnostic Extensions**: Emerging directions in dataset-level bias profiling, contextual-intersectional stress testing, and continuous attribute partitioning.

Table 2-1 at the conclusion of this chapter consolidates these twenty contributing works, highlighting their specific architectural tier and operational role within BiasAperture.

2.1. The Empirical Mandate for Facial Analysis Bias Auditing

Buolamwini and Gebru's *Gender Shades* study audited commercial gender-classification APIs across skin-type and gender intersections [1]. They reported that aggregate accuracy masked subgroup differences: error rates ranged from 0.8% for lighter-skinned male subjects to 34.7% for darker-skinned female subjects. The study motivates reporting subgroup results alongside aggregate measures. The proposed BiasAperture workflow includes intersectional cohorts based on the FairFace demographic categories; this design does not imply that every combination is adequately represented or measured.

Stanley et al. studied bias encoding in convolutional networks using controlled synthetic medical images [9]. Their work motivates caution about interpreting subgroup differences as evidence of a particular mechanism. The demographic logistic surrogate reviewed for BiasAperture uses race, gender, and age indicators to predict model outputs. It summarizes associations between those demographic indicators and predictions; it does not analyse image pixels, determine whether a visual feature is a proxy, identify confounders, or establish the cause of a disparity.

Nemavhola et al. evaluated ResNet-50, MobileNetV3, and Vision Transformer (DeiT) architectures on FairFace and UTKFace [10]. They reported larger racial than gender disparities in their evaluated models, with reported effect sizes above $d = 1.0$. These are findings from that study, not results of the BiasAperture case study. UTKFace was excluded from BiasAperture's completed scope.

2.2. Mathematical Formulations of Group Fairness and Disparity

Hardt et al. introduced two widely used error-rate parity criteria in supervised learning: *Equalized Odds* and *Equal Opportunity* [11]. Equalized Odds requires equal True Positive Rates (TPR) and equal False Positive Rates (FPR) across demographic cohorts, conditional on the true label:

$$P(\hat{Y} = 1 \mid A = a, Y = y) = P(\hat{Y} = 1 \mid A = b, Y = y), \quad \forall y \in \{0, 1\}, \forall a, b. \tag{2-1}$$

Equal Opportunity requires parity in TPR for the outcome defined as advantageous ($Y = 1$). These criteria address different questions. The proposal specifies Equalized Odds Difference (EOD) and Equal Opportunity Difference (EOP) as two of its four metrics; backend agreement is a computational cross-check, not independent evaluation.

Watkins et al. analysed the use of the U.S. Equal Employment Opportunity Commission's four-fifths rule in machine learning fairness [12]. Their discussion cautions against treating a ratio threshold as a context-free pass/fail test. The proposal accordingly specifies that Disparate Impact Ratio (DIR) and Demographic Parity Difference (DPD) be presented with statistical context rather than as standalone threshold decisions.

Howard et al. evaluated face-recognition fairness measures against functional criteria, including interpretability, boundedness, and behavior at zero-error cases [13]. Their analysis illustrates that metric choice can affect comparative conclusions. BiasAperture proposes separate Fairlearn and AIF360 implementations to cross-check calculations; this does not establish independent expert assessment or validate every methodological choice. Howard et al.'s work addresses biometric verification error rates, whereas BiasAperture's proposed metrics concern classification disparities.

2.3. Statistical Rigor, Uncertainty, and Demographic Inference Epistemology

A point estimate alone does not describe sampling uncertainty. Nemavhola et al. paired disparity measures with bootstrap confidence intervals, analysis of variance, and Cohen's d effect sizes [10]. In the reviewed implementation, subgroup confidence intervals are empirical percentile intervals from a bootstrap routine whose default is $B = 200$; the backend adapters do not pass a larger resample budget. This falls short of the proposed NFR-002 target of at least 1,000 resamples. A separate global-summary routine uses a stratified BCa bootstrap, but that is not the method or budget used for subgroup rows. These implementation details should accompany any interpretation of reported intervals.

Statistical reliability depends on adequate sample support. Small subgroup samples can yield unstable estimates. The implementation applies a minimum-size guard at $n < 30$, marking such rows insufficient rather than reporting a metric value.

Demographic auditing also depends on the provenance and reliability of demographic labels. Fournier-Montgieux et al. analyse how demographic-attribute-inference errors affect fairness-audit validity and report limitations in FairFace's published training pipeline and classifier performance relative to alternative inference heads [14].

Karkkainen and Joo's FairFace dataset has 97,698 released images and seven race, two gender, and nine age categories [8]. FairFace is the retained benchmark in the completed scope; UTKFace was excluded. In line with Fournier-Montgieux et al.'s discussion of demographic-inference error, FairFace labels should be treated as benchmark annotations rather than infallible ground truth. The report should record this limitation where results are presented.

2.4. Explainability Foundations and Bounds on Feature Attribution

Feature attribution methods draw on cooperative game theory. Lloyd Shapley established a payoff allocation for an n-player cooperative game that satisfies four axioms: *Efficiency*, *Symmetry*, *Dummy/Null Player*, and *Additivity* [15]:

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} \left[v(S \cup \{i\}) - v(S) \right]. \tag{2-2}$$

Lundberg and Lee proposed SHAP (SHapley Additive exPlanations), an additive feature-attribution framework, and derived a Shapley kernel for weighted linear regression [16]:

$$\pi_x(z') = \frac{M - 1}{\binom{M}{|z'|} \cdot |z'| \cdot (M - |z'|)}. \tag{2-3}$$

The reviewed surrogate implementation fits logistic regression on demographic indicator variables (race, gender, and age) to predict the audited model's predictions, then summarizes additive feature attributions. This is association-based attribution over demographic variables, not image explanation. It does not discover causes, identify proxy confounders, or decompose the measured disparity. Image-space attribution and Individual Typology Angle (ITA) analysis are outside the current scope.

Bilodeau et al. discuss limits on feature attribution's ability to distinguish causal drivers from correlations [17]; Slack et al. describe ways post-hoc explainers can be fooled [18]. These findings motivate bounded interpretation. Demographic-surrogate output should be reported as association-based attribution, not as image explanation, causal discovery, proxy-confound identification, disparity decomposition, or evidence that bias is absent.

2.5. Standardized Documentation, Governance, and Regulatory Context

A diagnostic audit benefits from clear reporting for technical and non-technical readers. The proposed report structure draws on established documentation conventions and regulatory sources.

Mitchell et al. proposed *Model Cards for Model Reporting*, a documentation format for a model's intended domain, performance, evaluation datasets, and ethical considerations [6]. Gebru et al. proposed *Datasheets for Datasets*, a framework for documenting dataset provenance, collection, annotation, and limitations [7]. The planned HTML report is informed by these conventions and would describe the FairFace benchmark and the exclusion of UTKFace from completed scope.

Buscemi et al. discuss decomposing high-risk requirements in the European Union Artificial Intelligence Act (Regulation (EU) 2024/1689, as amended by Regulation (EU) 2026/1744) into technical verification activities [3, 4, 19]. The proposed BiasAperture crosswalk (Section 4.5) is a research aid, not a clause-level compliance assessment. Article 10(2)(f) concerns examining datasets for possible bias; paragraph 10(2)(g) concerns measures to detect, prevent, and mitigate biases; paragraph 10(3) addresses dataset quality and representativeness; and paragraph 10(4) addresses relevant geographical, contextual, behavioural, or functional settings. Regulation (EU) 2026/1744 deleted Article 10(5) and inserted Article 4a, which provides a conditional legal basis for strictly necessary processing of special categories of personal data for bias detection and correction, subject to safeguards and conditions [4].

The proposed diagnostic controls can be discussed in relation to the *Measure* function of the NIST AI RMF 1.0 [5]. Measure 2.11 addresses evaluation of fairness and bias; Measure 1.1 concerns documented approaches for measuring AI risks. Measure 1.3 calls

for regular assessment involving internal experts not involved in system development or independent assessors. Fairlearn and AIF360 are software backends, not independent assessors, and their comparison does not satisfy Measure 1.3.

2.6. Reusable Pipelines for Computer Vision Auditing

Dehdashtian et al. survey fairness and bias mitigation across the computer-vision lifecycle, including data collection, curation, annotation, training, and post-deployment monitoring [2]. Their review describes a broad literature on metrics and mitigation methods. BiasAperture addresses a narrower engineering goal: specifying a reusable workflow for subgroup evaluation of facial-analysis predictions. General-purpose fairness toolkits such as Fairlearn and AIF360 may require data preparation to apply to this setting.

The proposed BiasAperture workflow is intended to combine dataset ingestion, fairness calculations, statistical summaries, bounded demographic-surrogate attribution, and HTML reporting. It does not introduce a new fairness metric or mitigation method. Reports are intended as traceable research evidence, not compliance reports or proof of compliance. The design excludes model retraining, weight modification, and synthetic augmentation.

2.7. Frontiers and Future Diagnostic Extensions

This section reviews three works relevant to possible extensions beyond the current scope.

First, Cascone et al. established a formal framework for evaluating representational and stereotypical bias at the dataset level, prior to and independent of model training [20]. Auditing CelebA, LFW, and MAAD-Face, the authors demonstrated that representational disparities collapse to a single latent factor, whereas stereotypical attribute correlations vary widely across datasets. This work motivates a future **Tier 2** extension for BiasAperture: an upstream dataset-profiling module that audits training and validation corpora for representational skew and stereotypical correlation before model predictions are ingested.

Second, Aslam et al. introduced *CIFA* (Contextual-Intersectional Fairness Auditing), demonstrating that traditional demographic-only audits fail to uncover severe "hidden subgroup" performance degradations caused by environmental and contextual factors [21]. Evaluating models across combined demographic and contextual axes (such as illumination variations, motion blur, and facial accessories), CIFA revealed performance drops of over 26% in contextual subgroups that aggregate audits completely masked. While CIFA's proposed mitigation re-training is out of scope for BiasAperture's

diagnostic mandate, its contextual-intersectional evaluation paradigm represents a direct candidate for future diagnostic stress testing.

Finally, Shilova et al. addressed the limitations of fixed categorical demographic schemas in their *FairGroups* framework [22]. The authors demonstrated that quantizing continuous biological attributes—such as skin tone represented in continuous CIELAB color space—into coarse, predefined discrete bins (such as the 6-point Fitzpatrick scale) can obscure substantial intra-group disparities. Using data-driven partitioning, FairGroups identified vulnerable subgroups that standard discrete bins missed. For BiasAperture, whose Milestone M1 schema is locked to discrete categorical axes ($7 \times 2 \times 9$), Shilova et al.'s work provides the theoretical blueprint for a future continuous-attribute auditing module that evaluates continuous skin reflectance (ITA) alongside discrete demographic taxonomies.

2.8. Summary of Reviewed Literature

Table 2-1 consolidates the twenty works reviewed in this chapter, summarizing their contributions and relevance to the proposed BiasAperture design.

Table 2-1. Literature review matrix: 20 works and their relevance to the proposed BiasAperture design.

Authors	**Paper Title (Year)**	**Contribution**	**Relevance to Proposed Design**
Theme 1: Empirical Foundations of Intersectional Bias			
Buolamwini & Gebru [1]	Gender Shades (2018)	Audits commercial facial analysis APIs; reveals error rates up to 34.7% on darker females vs 0.8% on lighter males.	**Core Foundation (Tier 1).** Motivated automated intersectional cohort stratification ($7 \times 2 \times 9$) over aggregate accuracy metrics.
Stanley et al. [9]	Bias Encoding in CNNs (2025)	Studies bias encoding in synthetic medical images, including associations with texture, lighting, and other visual features.	**Supporting Literature (Tier 1).** Motivates caution about mechanisms; BiasAperture's demographic surrogate does not identify visual proxies.

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Authors	Paper Title (Year)	Key Contribution	Architectural Tier & Adopted Concepts
Nemavhola et al. [10]	Reassessing Demographic Bias (2026)	Multi-model audit on FairFace and UTKFace; reports larger racial than gender disparities ($d > 1.0$) in evaluated models.	**Supporting Literature (Tier 1).** Provides findings from a separate study for context; these are not BiasAperture case-study results.
Theme 2: Mathematical Formulations of Fairness			
Hardt et al. [11]	Equality of Opportunity (2016)	Formulates Equalized Odds (TPR+FPR parity) and Equal Opportunity (TPR parity) in supervised learning.	**Core Foundation (Tier 1).** Directly adopted as the mathematical definitions of EOD and EOP in the Core Four metric battery.
Watkins et al. [12]	Four-Fifths Rule Critique (2022)	Critiques importing employment law thresholds (80%) into ML as epistemically unjustified bare ratios.	**Core Foundation (Tier 1).** Governs NFR-001: DIR is never reported as a bare ratio; always paired with χ^2 testing and bootstrap CIs.
Howard et al. [13]	Evaluating Proposed Fairness Models (2022)	Evaluates FR fairness metrics against FFMC criteria; demonstrates metric instability and ranking inversions.	**Supporting Literature (Tier 1).** Informs comparison of Fairlearn and AIF360 calculations; this is a software cross-check, not independent assessment.
Theme 3: Statistical Rigor & Demographic Epistemology			
Nemavhola et al. [10]	Statistical Grounding in Bias (2026)	Reports bootstrap intervals and effect sizes alongside disparity estimates.	**Core Foundation (Tier 1).** Informs the proposed uncertainty reporting; the reviewed subgroup implementation currently uses percentile intervals with default $B = 200$, below NFR-002's target.
Fournier-			

Continued on next page

Authors	Paper Title (Year)	Key Contribution	Architectural Tier & Adopted Concepts
Montgieux et al. [14]	Reliable Demographic Inference (2025)	Analyses how demographic-inference quality affects fairness-audit validity and discusses FairFace reproducibility limitations.	**Validity Consideration (Tier 1).** Supports treating FairFace annotations as benchmark labels rather than infallible ground truth.
Karkkainen & Joo [8]	FairFace Benchmark (2021)	Constructs balanced 97,698-image face dataset with 7-race, 2-gender, and 9-age categories to minimize skew.	**Core Benchmark (Tier 1).** Primary validation dataset (FR-001); UTKFace excluded due to synthetic age noise and race collapsing.
Theme 4: Explainability Foundations & Theoretical Bounds			
Shapley [15]	Value for n-Person Games (1953)	Axiomatic cooperative game theory defining the Shapley value through four axioms.	**XAI Foundation (Tier 1).** Background for feature attribution; it does not establish causal interpretation of attributions.
Lundberg & Lee [16]	Unified Approach to Interpretation (2017)	Presents additive feature attribution (SHAP) and a Shapley kernel for weighted linear regression.	**XAI Foundation (Tier 1).** Related attribution literature; the reviewed logistic demographic surrogate is not image-based SHAP.
Bilodeau et al. [17]	Impossibility Theorems (2022)	Analyzes limits on attribution methods' ability to distinguish causal drivers from correlations.	**XAI Theoretical Bound (Tier 1).** Supports reporting surrogate output as association-based attribution, not causal evidence.

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Authors	Paper Title (Year)	Key Contribution	Architectural Tier & Adopted Concepts
Slack et al. [18]	Fooling LIME and SHAP (2020)	Demonstrates adversarial scaffolding can disguise discriminatory model behavior from post-hoc explainers (grounds surrogate attribution boundary).	**XAI Threat Model (Tier 1).** Bounds trust in post-hoc XAI; motivates dual model interfacing (predictions file vs in-process inference).
Theme 5: Standardized Documentation & Governance			
Mitchell et al. [6]	Model Cards (2019)	Proposes standardized, stakeholder-legible documentation format for model performance and intended use.	**Core Reporting (Tier 1).** Adopted as structural foundation for the model performance evaluation section of generated HTML reports.
Gebru et al. [7]	Datasheets for Datasets (2018)	Proposes structured documentation for dataset provenance, collection mechanisms, and demographic composition.	**Core Reporting (Tier 1).** Informs the proposed dataset-provenance section and its scope notes.
Buscemi et al. [19]	Assessing High-Risk AI Systems (2025)	Decomposes EU AI Act high-risk obligations into concrete, technology-agnostic technical verification activities.	**Regulatory Crosswalk (Tier 1).** Informs a proposed research crosswalk to relevant AI Act provisions; it is not a compliance assessment (Section 4.5).
NIST [5]	AI Risk Management Framework (2023)	Consensus risk management taxonomy organizing AI governance into Govern, Map, Measure, and Manage functions.	**Regulatory Crosswalk (Tier 1).** Informs proposed alignment to Measure 2.11 and 1.1; Measure 1.3 requires independent assessors or uninvolved internal experts, not dual software backends.

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Authors	Paper Title (Year)	Key Contribution	Architectural Tier & Adopted Concepts
Theme 6: Pipeline Engineering & Diagnostic Tooling			
Dehdashtian et al. [2]	Fairness in CV: A Survey (2024)	Surveys bias sources across CV pipeline; reveals rich mitigation literature but immature reusable auditing pipelines.	**Workflow Gap (Tier 1).** Informs the proposed reusable diagnostic workflow; implementation and validation remain separate evidence questions.
Theme 7: Frontiers & Future Diagnostic Extensions			
Cascone et al. [20]	Bias-Aware Dataset Evaluation (2026)	Taxonomy distinguishing model-output bias from dataset-level representational and stereotypical attribute bias.	**Future Extension (Tier 2).** Motivates candidate upstream dataset-profiling module prior to and distinct from model inference audits.
Aslam et al. [21]	CIFA: Contextual Auditing (2026)	Demonstrates demographic audits mask severe errors under contextual variations (illumination, blur, accessories).	**Future Extension (Tier 2).** Informs future contextual-intersectional stress testing layer; mitigation steps excluded per diagnostic scope.
Shilova et al. [22]	FairGroups: Continuous Variables (2025)	Data-driven partitioning of continuous sensitive attributes (skin tone in CIELAB) to expose hidden intra-group bias.	**Future Extension (Tier 2).** Theoretical basis for future continuous skin-tone (ITA) evaluation layer on top of locked discrete M1 schema.

These twenty works inform the proposal's empirical, statistical, explainability, governance, and engineering choices. Chapters 3 and 4 set out corresponding requirements and an architectural design; neither the literature review nor the crosswalk establishes regulatory compliance.

3. REQUIREMENT ANALYSIS

This chapter records proposed functional and non-functional requirements for BiasAperture. Hardware and software requirements describe the intended build and run environment. They are acceptance targets, not evidence that every requirement has been implemented or validated; Chapter 4 describes the proposed architecture and verification plan.

3.1. Functional Requirements

Functional requirements are listed using IEEE 830-style "shall" statements, one capability per requirement.

1. **FR-001 (Data Ingestion):** The system shall load, validate, and standardise the FairFace benchmark dataset or a compatible custom dataset, checking image integrity and aligning demographic annotation fields (race or ethnicity, perceived gender, age group) into a common internal schema.
2. **FR-002 (Dual-Mode Model Interface):** The system shall obtain predictions from a target facial-analysis model either by direct in-process inference against a supplied PyTorch or TensorFlow model object, or by batch ingestion of a precomputed predictions file in Comma-Separated Values (CSV) or JavaScript Object Notation (JSON) format.
3. **FR-003 (Fairness Metric Computation):** The system shall compute, per subgroup and per demographic intersection, the four Core Four disparity metrics, DPD, EOD, EOP, and DIR, using separate AIF360 and Fairlearn implementations as computational cross-checks, not as independent expert assessments.
4. **FR-004 (Statistical Significance Testing):** The system shall accompany every reported disparity with a chi-squared test of independence between the predicted-label distribution and the demographic subgroup, and with a bootstrap confidence interval for the reported disparity estimate.
5. **FR-005 (Explainability):** The system shall generate demographic-dummy logistic-surrogate attribution (with image-based spatial SHapley Additive exPlanations (image-based spatial explanations deferred) (SHAP) analysis deferred) describing associations between demographic indicators and model predictions. This output shall not be presented as image explanation, causal discovery, proxy-confound identification, or disparity decomposition.
6. **FR-006 (Report Generation):** The system shall assemble the computed metrics, statistical tests, and dataset and model provenance information into a standardised, exportable HTML fairness report, structurally informed by the Model Cards and Datasheets for Datasets conventions, with every metric row showing subgroup

sample size, point estimate, 95% confidence interval, and either a p-value or an insufficient-sample flag.

7. **FR-007 (Regulatory Traceability):** The proposed report shall map relevant metrics to cited provisions of Article 10 or Annex IV of the EU AI Act and to relevant outcomes in the NIST AI RMF, with explanatory caveats. This crosswalk is traceable research evidence, not a legal assessment or proof of compliance.
8. **FR-008 (Orchestration and Configuration):** The system shall expose a command-line configuration interface through which an auditor selects the benchmark dataset, the model or predictions source, and the metrics to compute, and shall re-run identically given the same configuration and inputs.
9. **FR-009 (Licensing Acknowledgement):** The system shall display and require explicit acknowledgement of the selected dataset's licensing terms before an audit is executed.
10. **FR-010 (Regulatory Evidence Traceability):** For every audit run and every reported metric, the system shall produce a machine-readable evidence manifest (manifest ID, timestamp, code version, dependency lockfile, dataset ID/checksum/licence, model ID and mode, CLI config, random seeds, and computed metrics with sample sizes) and embed it in the exported report bundle as `manifest.json`.
11. **FR-011 (Calibration and Threshold Audits):** The system shall support computing per-group calibration curves, expected calibration error (ECE), and thresholded metrics (precision@k, recall at a fixed false positive rate), allowing auditors to request thresholded disparity tables and visualize calibration by subgroup.

3.2. Non-Functional Requirements

1. **NFR-001 (Statistical Rigour):** All significance testing shall use a threshold of $\alpha = 0.05$, and the system shall report exact p-values rather than only a pass/fail flag.
2. **NFR-002 (Uncertainty Quantification):** Every reported subgroup disparity estimate shall include a 95% confidence interval computed from a minimum of 1,000 bootstrap resamples.
3. **NFR-003 (Data-Integrity Guard):** Any subgroup with a sample size below $n = 30$ shall be automatically flagged as "insufficient sample, not reported" rather than assigned a computed metric value.
4. **NFR-004 (Performance):** A full-dataset evaluation over FairFace's 97,698 released images (86,744 train + 10,954 validation) shall complete within 4 hours on a single mid-range Graphics Processing Unit (GPU); a stratified development subset of $n = 5{,}000$ images shall complete within 30 minutes on Central Processing Unit (CPU) alone.

5. **NFR-005 (Modularity):** The system shall maintain strict separation of concerns across its architectural modules, each exposing a narrow, well-defined interface, so that any module can be unit-tested and extended independently of the others.
6. **NFR-006 (Reproducibility):** Bootstrap resampling shall use fixed random seeds, and third-party dependency versions (AIF360, Fairlearn, PyTorch or TensorFlow) shall be pinned in a lock-file to prevent silent behavioural drift between runs.
7. **NFR-007 (Portability):** The system shall run cross-platform on any environment supporting the pinned Python dependency stack, without requiring proprietary or platform-specific components.
8. **NFR-008 (Explainability Performance):** Surrogate feature attribution (with spatial SHAP analysis deferred) shall be computed only on flagged disparities rather than the full dataset by default, to keep explainability overhead a small fraction of total runtime.
9. **NFR-009 (Reproducibility and Report Integrity):** Each report bundle shall include the code version (git commit SHA or container digest), a pinned dependency lockfile, CLI configuration, random seeds, `manifest.json`, and an integrity hash (SHA-256) for the report bundle, with optional HMAC signature support.
10. **NFR-010 (Statistical Power Guidance):** The system shall provide an on-demand power-analysis utility estimating minimum detectable effect size given subgroup sample sizes, significance level α, and statistical power $(1 - \beta)$, presenting sensitivity curves for small subgroups.

NFR-002 is a proposed target, not a claim of current conformance. In the reviewed implementation, subgroup confidence intervals use the percentile bootstrap with default $B = 200$; the adapters do not pass a larger budget. A separate global-summary path calls the BCa bootstrap routine, but that does not satisfy the subgroup requirement.

3.3. System Requirements

This section answers what machine and software stack are required to build and run BiasAperture.

3.3.1. Hardware Requirements

- **Minimum (development, stratified subset):** 64-bit CPU, 4 or more cores, 8 GB Random-Access Memory (RAM), 5 GB free disk space.
- **Recommended (full-dataset audit, NFR-004 target):** A single mid-range GPU with 8 GB or more Video Random-Access Memory (VRAM), for example an NVIDIA T4-class device, either a local workstation GPU or a free-tier cloud notebook GPU such as Google Colab.

- No specialised or custom hardware is required; the platform performs inference and metric computation only, never model training from scratch.

3.3.2. Software Requirements

- **Operating system:** Linux, Windows, or macOS; the stack is pure Python and cross-platform.
- **Language and runtime:** Python 3.10 or later.
- **Model interface:** PyTorch or TensorFlow, whichever framework the audited model was built in.
- **Image preprocessing:** OpenCV, pandas, NumPy.
- **Fairness computation:** AIF360, Fairlearn, scikit-learn, SciPy.
- **Explainability:** Demographic-dummy surrogate attribution (scikit-learn; SHAP deferred).
- **Report generation:** Jinja2 for HTML templating.

4. SYSTEM ARCHITECTURE AND METHODOLOGY

Chapter 3 specified proposed system requirements. This chapter describes the proposed architecture, regulatory research crosswalk, project plan, and verification strategy. Design descriptions and planned checks are not evidence that every requirement has been implemented or validated.

4.1. Design Principles

The proposed architecture uses object-oriented design concepts and the Single Responsibility, Open-Closed, Liskov Substitution, Interface Segregation, Dependency Inversion (SOLID) principles as guidance. Its five-module design separates ingestion, model access, metric computation, attribution, and reporting (Section 4.2); strategy and adapter patterns are proposed at the fairness and model interfaces (Section 4.4). These design descriptions do not establish that interfaces are interchangeable or that all principles have been verified in the implementation. The shared schema and parallel work streams are part of the original project plan (Section 4.7), not evidence of completed parallel development.

4.2. Architectural Modules

This section presents the proposed system architecture. Its modules describe intended responsibilities and interfaces; Fig. 4-1 shows the planned flow from dataset and model inputs to research-report outputs.

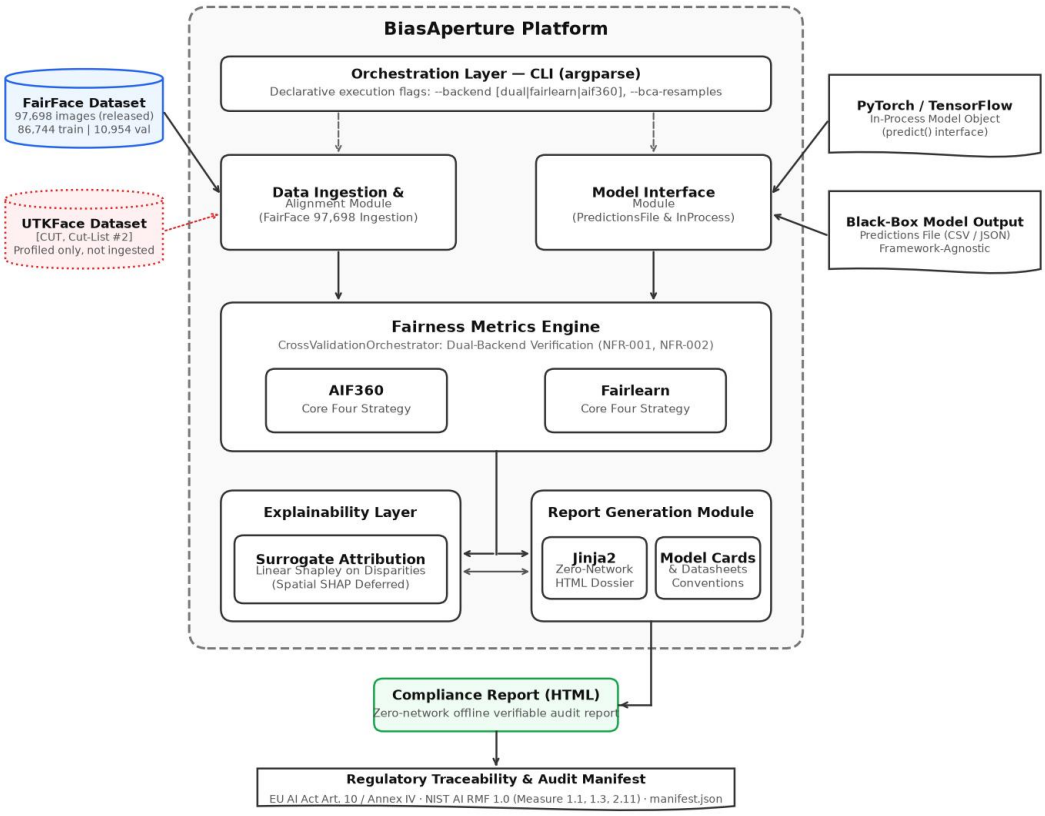


Figure 4-1. BiasAperture system architecture.

Table 4-1 summarizes proposed module responsibilities, related requirements, and intended libraries. The following subsections describe their planned roles.

Table 4-1. Architectural modules and their responsibilities.

Proposed Module	**Intended Responsibility**	**Planned Libraries**
Data Ingestion and Preprocessing	Load, validate, and standardise a benchmark or custom dataset (FR-001)	OpenCV, pandas, NumPy
Model Interface	Obtain predictions via direct inference or a predictions file (FR-002)	PyTorch, TensorFlow
Fairness Metrics Engine	Compute subgroup and intersectional metrics with significance testing (FR-003, FR-004)	AIF360, Fairlearn, scikit-learn, SciPy
Explainability Layer	Summarize demographic-indicator associations with model predictions (FR-005)	Logistic surrogate; image-based SHAP deferred
Report Generation	Assemble traceable research evidence in a Model-Cards- and Datasheets-informed report (FR-006, FR-007)	Jinja2
Orchestration and Configuration	Coordinate the pipeline end to end behind a single entry point (FR-008, FR-009)	Python Command-Line Interface (CLI)

4.2.1. Data Ingestion and Preprocessing Module

This module is specified to load a selected benchmark or compatible custom dataset, validate image integrity, standardise image resolution and colour space via OpenCV, and align demographic annotation fields (race or ethnicity, perceived gender, age group) into the proposed internal schema (Section 4.7). FairFace is the retained benchmark; UTKFace was excluded from completed scope.

4.2.2. Model Interface Module

The proposed model interface has two modes: direct in-process inference against a supplied PyTorch or TensorFlow model exposing a standard predict interface, and batch ingestion of a precomputed predictions file in CSV or JSON format. Neither mode requires changing model weights; training and fine-tuning are out of scope.

4.2.3. Fairness Metrics Computation Engine

The proposed analytical core computes subgroup and intersectional performance and disparity metrics using AIF360 and Fairlearn. Comparing the backends can expose implementation differences, but does not constitute independent expert assessment. The implementation also has two distinct interval paths: the subgroup helper defaults to empirical percentile intervals with $B = 200$, and the adapters do not pass a higher budget; a separate global-summary path calls a configurable BCa bootstrap function. The subgroup default therefore falls short of proposed NFR-002's minimum $B = 1{,}000$. The proposal also specifies significance testing.

4.2.4. Explainability Layer

The reviewed logistic surrogate uses demographic indicator variables (race, gender, and age) to predict model outputs and summarizes associations. It does not operate on image pixels and cannot determine whether lighting, background, or another visual feature is a proxy or confounder. It does not establish causation or decompose the measured disparity. Image-based spatial SHAP remains out of scope.

4.2.5. Report Generation Module

The proposed module assembles metrics, tests, and dataset and model provenance into a Model-Cards- and Datasheets-informed HTML report. FR-006 specifies subgroup size, point estimate, confidence interval, and either a p-value or insufficient-sample flag. The current subgroup interval default is $B = 200$ percentile resamples, not the NFR-002 target of at least 1,000. The proposed Article 10, Annex IV, and NIST references are research traceability and do not certify compliance.

4.2.6. Orchestration and Configuration Layer

A lightweight layer coordinates the pipeline end to end, dataset selection, model connection, metric selection, and report generation, behind a single entry point, and exposes a command-line interface with declarative execution flags through which an auditor specifies the audit parameters (FR-008). The dataset's licensing terms must be displayed and explicitly acknowledged before an audit is executed (FR-009).

4.3. End-to-End Workflow of the System

Fig. 4-2 presents the proposed sequence for an audit run, from configuration through to a research report.

The proposed workflow begins with the auditor specifying the dataset, model or predictions file, and metrics. Its two model-access modes are direct in-process inference and ingestion of a precomputed file. The planned case study uses FairFace; UTKFace was excluded from completed scope.

The workflow specifies computation of the Core Four metrics, sample-size checks, statistical tests, and demographic-surrogate associations. The output is a research report, not a compliance report. Subgroup confidence intervals currently use the percentile bootstrap with default $B = 200$, which is below the proposed NFR-002 budget; these intervals are not BCa intervals and are not based on at least 1,000 resamples.

4.4. Design Patterns

Table 4-2 records patterns proposed for specific interfaces or scope decisions; it does not claim that each pattern has been implemented or tested.

4.5. Regulatory Traceability Mapping

FR-007 proposes a research crosswalk between metrics and regulatory sources. Table 4-3 summarizes relevant Article 10 provisions for discussion; it does not show that system outputs fulfil legal obligations. Article 10 timing depends on the applicable system category and the amended Article 113 timetable [3, 4, 19].

The proposed design can be discussed in relation to the NIST AI RMF's Govern, Map, Measure, and Manage functions; this alignment does not mean the framework's outcomes have been satisfied. The demographic schema relates to Map, and metric calculations relate to Measure. Measure 1.3 calls for regular assessment by internal experts not involved in development or by independent assessors. A comparison between Fairlearn and AIF360 is not independent assessment and does not meet that practice. Manage,

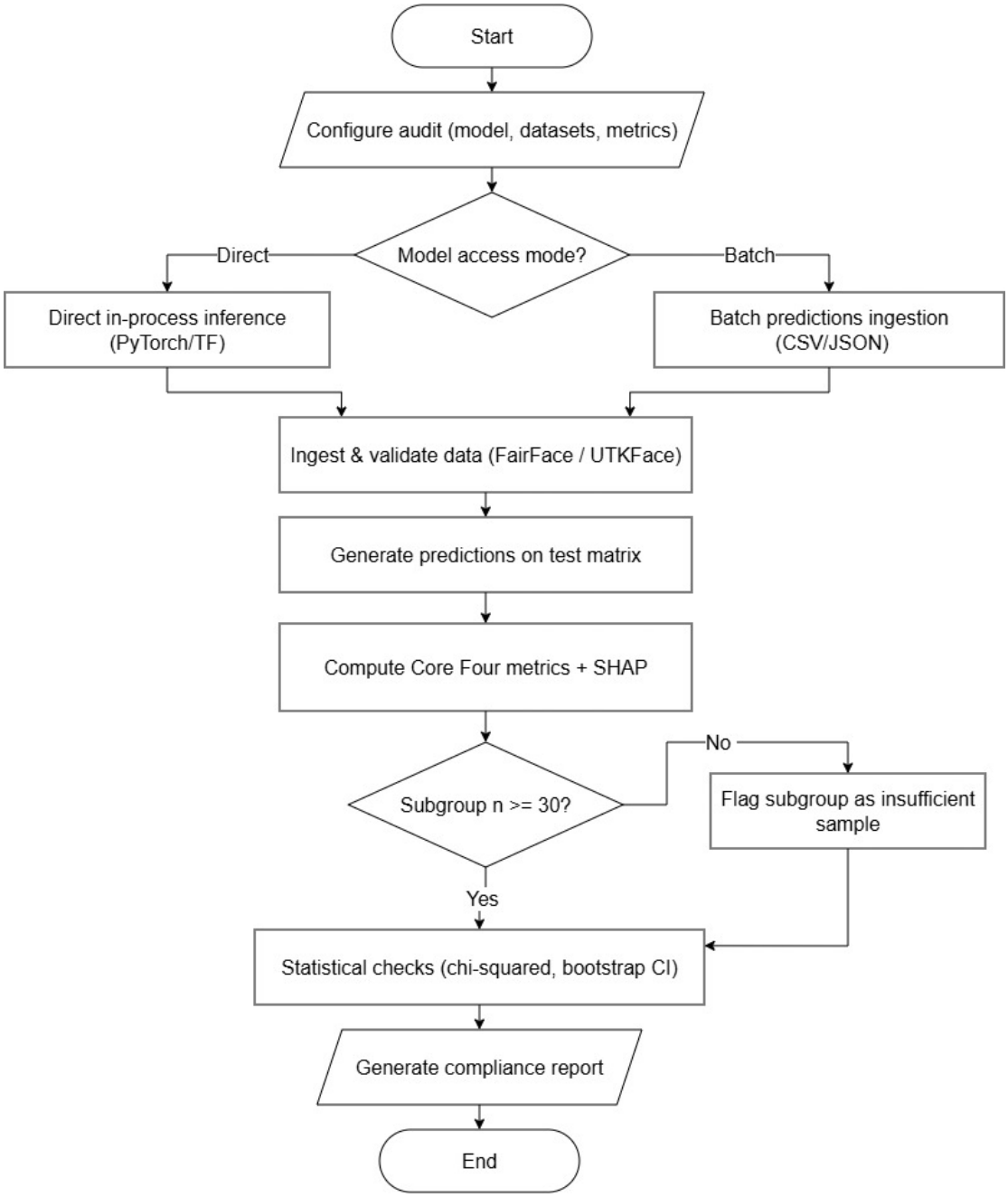


Figure 4-2. BiasAperture end-to-end audit workflow.

Table 4-2. Design patterns mapped to architectural modules.

Pattern	Module	Structure	Rationale
Adapter	Model Interface	`ModelInterface` → `InProcessInterface` / `PredictionsFileInterface`	Cutting direct inference (Section 4.8, item 4) means simply not instantiating one interface; downstream code is unchanged
Strategy	Fairness Metrics Engine	`FairnessBackend` → `AIF360Backend` / `FairlearnBackend`	Cutting AIF360 (Section 4.8, item 5) is a backend swap, not a rewrite; the four metrics are interchangeable strategies
Builder	Data Ingestion	`DataIngestionPipeline`	Matches the multi-step curation (acquisition, integrity check, schema alignment) the module performs
Factory	Report Generation	`HTMLReportGenerator`	Keeps the HTML report path untouched regardless of which export formats are in scope
Facade	Orchestration	`CrossValidationOrchestrator`	Hides the ingestion → interface → engine → report sequence behind one call

Table 4-3. Article 10 provisions and the limits of the proposed research crosswalk.

Article 10 Provision	**Requirement Summary**	**Potential Research Link and Limitation**
10(2)(f)–(h) Data governance	Examine possible bias, take appropriate detection/prevention/mitigation measures, and identify relevant data gaps	Proposed metrics and dataset documentation can inform research review; they do not establish fulfilment
10(3) Data quality	Relevance, representativeness, and quality in view of intended purpose	Dataset description and subgroup counts provide limited descriptive evidence, not a quality or compliance finding
10(4) Context	Account for relevant geographical, contextual, behavioural, or functional setting	Dataset provenance may document context; no completeness claim is made
Article 4a (inserted by Regulation (EU) 2026/1744)	Permits strictly necessary processing of special-category data for bias detection/correction in high-risk AI systems, subject to safeguards and conditions	BiasAperture's research metrics do not establish that these legal conditions are met; Article 10(5) was deleted by Regulation (EU) 2026/1744 [4]

including prioritization and response to identified risks, remains outside the diagnostic scope.

4.6. Work Breakdown Structure

Table 4-4 decomposes the project into deliverable-oriented work packages following the 100% rule, every module in Section 4.2 is covered and nothing is added outside stated scope, with leaf packages sized for a two-person team and each leaf tied either to an acceptance criterion in Chapter 3 or to a module in Section 4.2. Ownership is tagged *Together* where a package requires shared judgement or iteration, or *Stream A* / *Stream B* where it can proceed solo; named assignment of which team member takes which solo stream remains open at time of writing and will be settled by demonstrated preference rather than fixed in advance.

4.7. Project Plan and Schedule

The work packages in Table 4-4 record the original proposed eight-week schedule. They describe planned sequencing and ownership, not reported completion status.

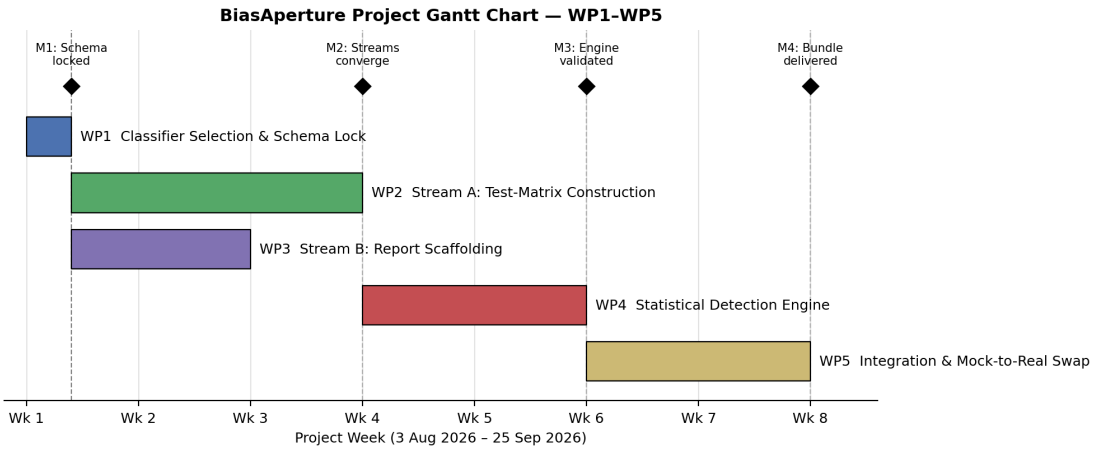


Figure 4-3. BiasAperture project schedule, work packages WP1–WP5.

WP1, *Classifier Selection and Schema Lock*, was planned to select the classifier baseline and define the demographic schema: `image_id`, race, gender, and age fields, true and predicted labels, and metric output fields. Its planned completion was Milestone M1, *schema locked*.

Under the plan, WP2 (Stream A, test-matrix construction) and WP3 (Stream B, report scaffolding) would proceed in parallel: Stream A would curate the FairFace-based dataset and predictions, while Stream B would build an HTML report template against a mock metrics dictionary. Their planned convergence was Milestone M2.

Table 4-4. Work breakdown structure.

ID	Work Package	Ownership
1.1	Project management: feasibility review, acceptance-criteria lock, schema lock, submission and defence prep	Together
1.2	Classifier selection and model interface: baseline confirmation, `ModelInterface` abstraction, predictions-file ingestion path	Together
1.3	Stream A, test-matrix construction: dataset acquisition and integrity check, inference run, schema curation, stratified development subset	Stream A
1.4	Stream B, report scaffolding: HTML and Jinja2 template, Model Cards / Datasheets structure, hand-written mock metrics dictionary	Stream B
1.5	Statistical detection engine: backend integration, four disparity metrics, chi-squared testing, bootstrap confidence interval, sample-size guard	Together
1.6	Integration: Stream A output into the detection engine, mock-to-real swap of Stream B's report against real findings, orchestration module	Together
1.7	Verification and validation: runtime targets, report-completeness rule, case-study run	Together
1.8	Documentation and delivery: scope-boundary statement, final report, submission package	Together

WP4, the *Statistical Detection Engine*, was planned to consume the schema-aligned test matrix and produce the metrics dictionary for the report template. Milestone M3 was defined as its planned end-to-end validation point.

WP5, *Integration and Mock-to-Real Swap*, was planned to connect engine output to the report template and complete with a case-study run and report bundle at Milestone M4.

4.8. Descoping Strategy

The scope in Chapter 3 records the original proposed plan. The descoping table preserves the planned order; it is not a statement that all listed capabilities remain future work.

Table 4-5. Descoping order if the project falls behind schedule.

#	Cut	Rationale
1	Web User Interface (UI), keep CLI only	Pure user-experience layer; no bearing on the fairness-engineering core
2	Exclude UTKFace; retain FairFace	UTKFace was excluded from completed scope; it is not part of the current case study
3	Portable Document Format (PDF) export, retain HTML	The proposal treated HTML as the initial report format
4	Direct in-process inference, retain predictions-file ingestion	A lower-complexity proposed fallback for the case study
5	AIF360, retain Fairlearn	A proposed fallback that would remove the second metric-computation backend; it would not remove the need for independent assessment

The table records the proposal's intended priorities; it is not evidence that any capability was completed or validated.

4.9. Verification and Validation Strategy

The verification strategy proposed for the project is to compare statistical routines with known reference values, validate data against the specified schema, and assess runtime and report completeness against the requirements. These are planned checks, not reported results in this chapter. In particular, no measured runtime or completed end-to-end case-study result is asserted here. NFR-002 remains unmet for subgroup intervals in the reviewed implementation because their percentile-bootstrap default is $B = 200$.

5. CONCLUSION

5.1. Summary

This report presents the proposal, requirements, and architecture for BiasAperture, a diagnostic platform for evaluating demographic accuracy disparities in third-party facial-analysis models. It reviews the fairness literature and regulatory sources, then describes a proposed workflow, a FairFace-only benchmark scope, and a research-oriented regulatory crosswalk. UTKFace was excluded from the completed scope. These design and literature-review contributions do not establish that the proposed platform has met every objective or regulatory obligation. The subgroup confidence-interval implementation uses percentile bootstrap with default $B = 200$, and adapters do not pass a larger budget; this falls short of proposed NFR-002's minimum of 1,000 resamples. The separate global BCa routine does not remedy that subgroup gap. Fairlearn and AIF360 are metric-computation backends, not independent assessors as described by NIST AI RMF Measure 1.3.

The proposed system is diagnostic: it reports measured disparities and uncertainty but does not remediate them, retrain a model, or generate synthetic demographic data. Its report is intended to provide traceable research evidence and must not be described as regulator-ready, a compliance dossier, or proof of compliance.

5.2. Future Work

The proposal's descoping order in Table 4-5 records planned options for the project. UTKFace was excluded from completed scope and is not part of the current case study; any future evaluation would require a separate decision and further review of label limitations. Other possible extensions include a web UI, PDF export, and additional model interfaces. As Dehdashtian et al. distinguish diagnostic evaluation from mitigation [2], any mitigation-facing work would also be a separate extension rather than a capability claimed here.

A. PROJECT BUDGET

BiasAperture is a pure-software diagnostic platform with no bill of materials or physical hardware deliverable (Chapter 3, System Requirements); the budget below is scoped accordingly to cloud compute, subscriptions, and documentation costs rather than component procurement.

A.1. Cloud Computing and Development Services

Table A-1. Cloud computing and development services cost.

Service	Description	Cost
Open-source software	Python, PyTorch/TensorFlow, AIF360, Fairlearn, scikit-learn, SciPy, OpenCV, scikit-learn demographic surrogate (spatial SHAP deferred), Jinja2	Free
Cloud GPU compute (NFR-004 target)	Google Colab, free-tier T4-class GPU for full-dataset evaluation runs	Free
Cloud GPU compute (contingency)	Google Colab Pro, if free-tier quota is exhausted during the full 97,698-image FairFace evaluation (NFR-004)	$10/month
Cloud storage	Google Drive, dataset and checkpoint storage during development	Free (existing quota)
	Subtotal	**$0–$10/month**

A.2. Documentation and Miscellaneous

Table A-2. Documentation and miscellaneous cost.

Category	Item / Description	Estimated Cost
Printing and binding	Proposal, progress, and final report copies for submission and defence	$10–$15
Miscellaneous	Contingency for unplanned documentation or minor tooling needs	$5–$10
	Total Estimated Cost	**$15–$35 one-time, plus $0–$10/month cloud compute contingency**

B. PROJECT TIMELINE

This appendix reproduces the original proposed eight-week schedule from Section 4.7, structured into work packages WP1–WP5 as detailed in Table 4-4. Figure B-1 shows the planned task durations, Stream A / Stream B concurrency, and milestones M1–M4; it is not a completion record.

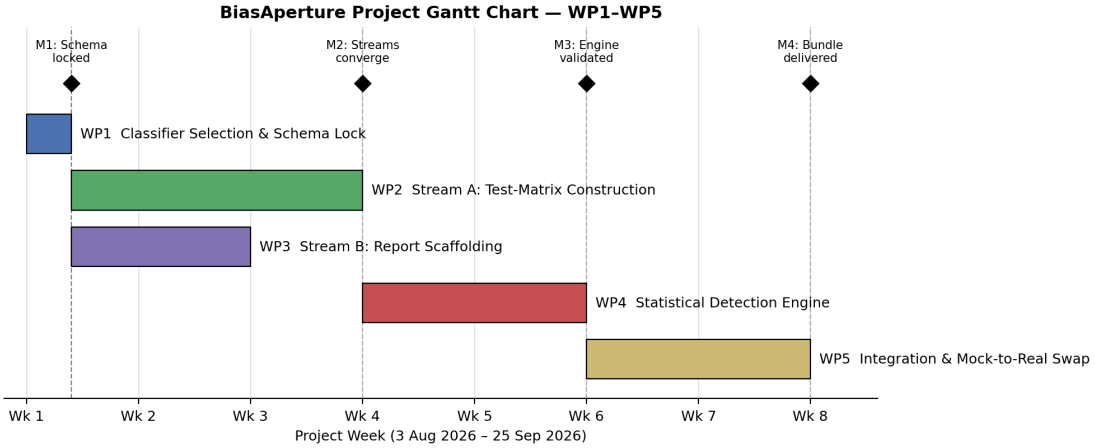


Figure B-1. BiasAperture project schedule, work packages WP1–WP5 (reproduced from Chapter 4).

C. LOCKED INTERNAL SCHEMA

Section 4.7 records the proposed internal schema for Stream A and Stream B. Tables C-1 and C-2 give its field-level specification.

Table C-1. Dataset / test-matrix schema.

Field	Type	Description
`image_id`	string	Unique identifier for the source image
`race`	categorical (7 levels)	FairFace race / ethnicity category
`gender`	categorical (binary)	Perceived gender label
`age`	categorical (9 bins)	FairFace age-group bin
`true_label`	categorical	Ground-truth demographic attribute value
`predicted_label`	categorical	Value returned by the Model Interface module
`protected_attribute`	categorical	Subgroup field currently selected for disparity analysis

Table C-2. Metrics-dictionary schema, the shape every Fairness Metrics Computation Engine output must satisfy.

Field	Type	Description
`metric_name`	string	One of DPD, EOD, EOP, DIR
`subgroup_n`	integer	Subgroup sample size
`point_estimate`	float	Computed metric value
`ci_lower`, `ci_upper`	float pair	95% confidence interval bounds; subgroup implementation uses percentile bootstrap, default $B = 200$, below proposed NFR-002 target of ≥1,000; separate global-summary path uses BCa
`p_value`	float	Result of the chi-squared significance test
`insufficient_sample_flag`	boolean	True when subgroup $n < 30$ (NFR-003)

D. RISK REGISTER

Table D-1 reproduces the risk register from the feasibility study underlying this proposal, covering the risks judged material enough to shape a design decision elsewhere in this report.

Table D-1. Project risk register: likelihood, impact, and mitigation strategy.

Risk	Likelihood	Impact	Mitigation
Demographic label noise, particularly UTKFace's model-estimated age labels	Medium	Medium	UTKFace was excluded from completed scope; use FairFace as the retained benchmark and document that its demographic annotations are not infallible ground truth
Model interface incompatibility with non-standard prediction formats	Medium	Medium	Provide the documented CSV / JSON predictions-file ingestion path as a fallback to direct in-process inference
Misinterpretation of metrics by non-specialist stakeholders	Medium	High	Plain-language explanation for every metric, with explicit caveats on statistical significance and sample size, in the generated report
Scope creep toward mitigation or model retraining	Medium	High	Written scope document reviewed with the supervisor; report generation restricted to diagnostic, non-prescriptive output only
Licensing non-compliance for a custom user-supplied dataset	Low	High	Display and require explicit acknowledgement of licensing terms before an audit executes (FR-009)
Computational resource constraints delaying full-dataset evaluation	Low	Medium	Stratified development subset (n = 5,000) supported for development; expected hardware runtimes documented (NFR-004)
Dependency version drift across AIF360, Fairlearn, and the model framework	Medium	Low	Dependency versions pinned in a lock-file; periodic re-validation against upstream releases (NFR-006)
Dual-use risk: findings used to exploit rather than remediate a model's weaknesses	Low	Medium	Intended, permitted use stated explicitly in project documentation; platform restricted to diagnostic reporting only

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